CSA5112 - Cryptography and Network Security

4. Hill Cipher (2×2)

Aim: To implement the Hill Cipher using a 2×2 key matrix.

Algorithm:

1. Start.
2. Read plaintext.
3. Convert letters into numerical values.
4. Multiply the plaintext matrix by the key matrix.
5. Apply modulo 26.
6. Convert numbers back into letters.
7. Display ciphertext.
8. Stop.

Code:

```
#include<stdio.h>

int main()

{

    int key[2][2]={{3,3},{2,5}};

    char text[3];

    int a,b,c,d;

    printf("Enter 2-letter Plain Text: ");

    scanf("%s",text);

    a=text[0]-'A';

    b=text[1]-'A';

    c=(key[0][0]*a+key[0][1]*b)%26;

    d=(key[1][0]*a+key[1][1]*b)%26;
```

```
    printf("Cipher Text: %c%c",c+'A',d+'A');

    return 0;

}
```

Sample Input:

```
Plain Text: HI
```

Sample Output:

```
Cipher Text: TC
```

Result:

The Hill Cipher algorithm was successfully implemented using a 2×2 key matrix, and the plaintext was encrypted successfully.